

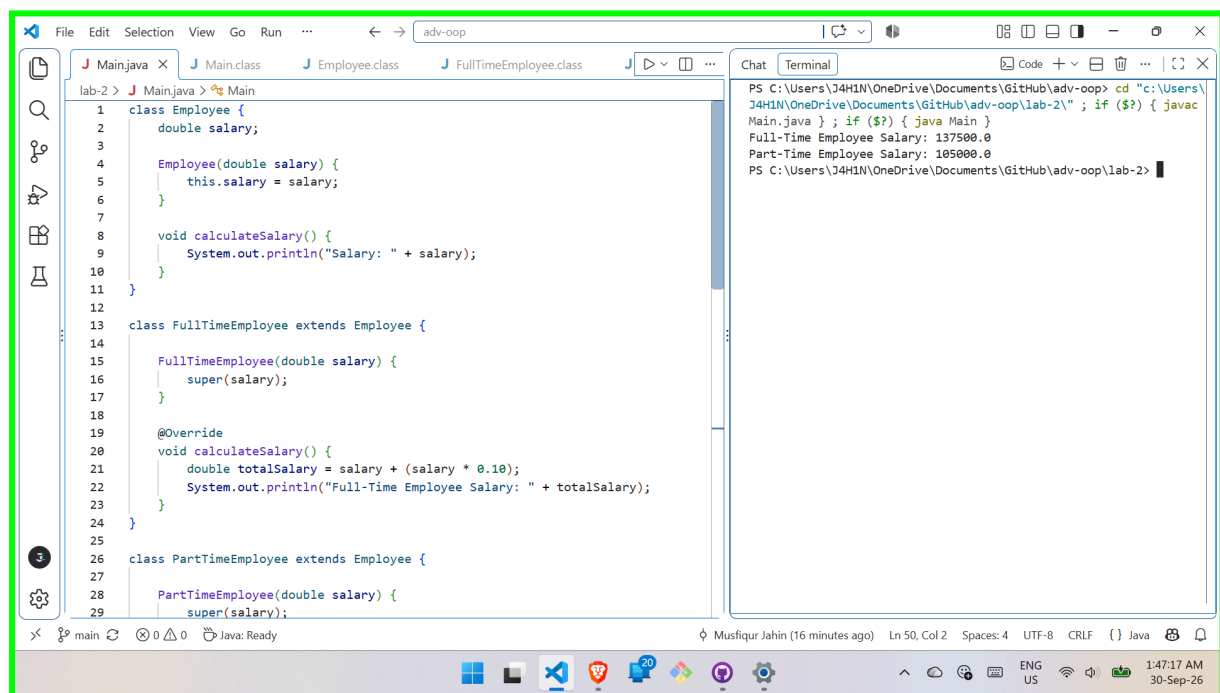
Experiment no : 02

Experiment name : Demonstration of Polymorphism Using Inheritance and Method Overriding in Java.

Problem Statement : Write a Java program to demonstrate polymorphism using inheritance and method overriding.

- Create a superclass "Employee" with a method "calculateSalary()".
- Create two subclasses, "FullTimeEmployee" and "PartTimeEmployee",
- Override the "calculateSalary()" method in each subclass according to their salary calculation.
- Use an "Employee" reference to call the overridden methods and display the salaries of both employees.

Code & Output:



The screenshot displays an IDE with a Java project named 'adv-oop'. The code editor shows three files: Main.java, Employee.class, and FullTimeEmployee.class. The code defines an 'Employee' superclass with a 'calculateSalary()' method that prints the salary. Two subclasses, 'FullTimeEmployee' and 'PartTimeEmployee', extend 'Employee' and override 'calculateSalary()' to calculate total salaries (including a 10% bonus for full-time employees). The 'Main' class creates instances of both subclasses and calls 'calculateSalary()' on them. The terminal output shows the execution results: 'Full-Time Employee Salary: 137500.0' and 'Part-Time Employee Salary: 105000.0'.

```
lab-2 > J Main.java > Main
1 class Employee {
2     double salary;
3
4     Employee(double salary) {
5         this.salary = salary;
6     }
7
8     void calculateSalary() {
9         System.out.println("Salary: " + salary);
10    }
11 }
12
13 class FullTimeEmployee extends Employee {
14
15     FullTimeEmployee(double salary) {
16         super(salary);
17     }
18
19     @Override
20     void calculateSalary() {
21         double totalSalary = salary + (salary * 0.10);
22         System.out.println("Full-Time Employee Salary: " + totalSalary);
23     }
24 }
25
26 class PartTimeEmployee extends Employee {
27
28     PartTimeEmployee(double salary) {
29         super(salary);
30     }
31 }
32
33 Main
34
35 PS C:\Users\J4H1N\OneDrive\Documents\GitHub\adv-oop> cd "c:\Users\J4H1N\OneDrive\Documents\GitHub\adv-oop\lab-2\" ; if ($?) { javac Main.java } ; if ($?) { java Main }
Full-Time Employee Salary: 137500.0
Part-Time Employee Salary: 105000.0
PS C:\Users\J4H1N\OneDrive\Documents\GitHub\adv-oop\lab-2>
```

Github Repo : { https://j4h1n0.web.app/?u=adv_oop_lab-2 }